produced by the explosion of a normal star. And the explosion was so hot, they suspected (this time incorrectly), that it radiated far more energy as ultraviolet light and X-rays than as ordinary light. Since the ultraviolet light and X-rays could not penetrate the Earth's atmosphere, it was impossible to measure just how much energy they contained. However, one could try to estimate their energy from the spectrum of the observed light and the laws of physics that govern the hot gas in the exploding supernova.

By combining Baade's knowledge of the observations and of ordinary novae with Zwicky's understanding of theoretical physics, Baade and Zwicky concluded (incorrectly) that the ultraviolet radiation and X-rays from a supernova must carry at least 10,000 and perhaps 10 million times more energy even than the visible light. Zwicky, with his love of extremes, quickly assumed that the larger factor, 10 million, was correct and quoted it with enthusiasm.

This (incorrect) factor of 10 million meant that during the several days that the supernova was at its brightest, it put out an enormous amount of energy: roughly a hundred times more energy than our Sun will radiate in heat and light during its entire 10-billion-year lifetime. This is about as much energy as one would obtain if one could convert a tenth of the mass of our Sun into pure, luminous energy!

(Thanks to decades of subsequent observational studies of supernovae —many of them by Zwicky himself—we now know that the Baade-Zwicky estimate of a supernova's energy was not far off the mark. However, we also know that their calculation of this energy was badly flawed: Almost all the outpouring energy is carried by particles called neutrinos and not by X-ray and ultraviolet radiations as they thought. Baade and Zwicky got the right answer purely by luck.)

What could be the origin of this enormous supernova energy? To explain it, Zwicky invented the neutron star.

Zwicky was interested in all branches of physics and astronomy, and he fancied himself a philosopher. He tried to link together all phenomena he encountered in what he later called a "morphological fashion." In 1932, the most popular of all topics in physics or astronomy was *nuclear physics*, the study of the nuclei of atoms. From there, Zwicky extracted the key ingredient for his neutron-star idea: the concept of a *neutron.*